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Lineage or Era? An Identifiability-Gated Instrument for Decomposing Correlated Errors in Public Language Models

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ABSTRACT Public language models make correlated errors: when one open-weight model fails a question, its contemporaries and descendants often fail it as well. Whether this shared failure structure is driven by lineage—a model replicating the blind spots of the models and data it descends from—or by era—models released together acquiring the same errors through shared training data and technique—cannot be settled by measurement alone, because the two are confounded by construction: descendants are released after their ancestors, making release date both a mediator and a confounder of the lineage path.

This paper contributes the instrument that such a measurement requires. We formulate the attribution as a crossed variance decomposition into lineage, era, and model-specific components, give sufficient conditions under which that partition is identifiable, and separate the observational estimand from a mechanistic one that holds era exactly fixed. We then validate the instrument in three gated stages, each completed before any trait value is observed. A structural audit establishes that the documented release record—not a reconstructed family tree—yields a crossed, connected 47-model design. A simulation study shows that a direct restricted-maximum-likelihood estimator recovers known ground truth to within 2.5 percentage points under balanced occupancy and 5.3 under the real one, and, critically, fails *detectably* when family and era are nested: three independent detectors flag the aliasing in 100% of repetitions with zero silent coverage. Finally, an outcome-independent design procedure, which never reads a trait value, selects the smallest population that remains structurally identifiable and clears a simulation-based recoverability bar—22 of 47 models, at roughly one-third the measurement cost of the full population. We show the structural minimum of 21 is insufficient, so the additional model is statistically necessary rather than merely convenient. The empirical shares are the object of a measurement protocol that this paper pre-registers in full; the contribution here is the gated instrument, and the demonstration that identifiability, estimator validity, and population design can be settled before measurement rather than defended after it.

INDEX TERMS Correlated errors, error correlation, foundation models, identifiability, mixed-effects models, restricted maximum likelihood, variance components, variance decomposition.

I. INTRODUCTION

THE growing reliance on a small number of public open-weight language models raises a statistical problem that is older than the models themselves: when a group of models is given a hard question, they often fail together—missing the same fact, making the same mistake, reaching the same wrong answer. Two explanations compete for this pattern of shared mistakes. The first is *lineage*: a model trained on another model's outputs, or on data generated by it, tends to replicate that model's blind spots. The second is *era*: models released at the same time are trained on largely the same scrapes of the web, the same benchmark-cleansed corpora, and the same dominant techniques, so they may pick up the same errors independently of any shared ancestry [1].

Whether shared failure is mostly inherited or mostly contemporaneous is not an academic question. If shared mistakes trace mostly to lineage, then diversification across model families is a credible mitigation: deploying models with independent ancestries reduces the probability that the ensemble fails a question the same way. If they trace mostly to era, then no amount of new ancestry helps, because every model of a given year carries the same errors. The choice between these two conclusions has direct operational consequences for how practitioners assemble model portfolios.

The consequences are concrete. Consider a review pipeline that routes each item to three open-weight models and escalates to a human only when they disagree. Its entire value rests on the assumption that the three fail independently enough that unanimous agreement is informative. If they do not—if a shared ancestor or a shared training year makes them wrong together on the same items—then unanimity is loudest exactly where it is least trustworthy, and the escalation rule silently stops firing on the cases that most need it. The operator cannot detect this from per-model accuracy, which may be excellent for all three. What matters is the structure of the shared component, and whether adding a fourth model from a different family reduces it. That is a question about variance components, not about accuracy, and it is the question this paper builds the instrument to answer.

The difficulty is that the two explanations are not separable by inspection. Descendants are always released after their ancestors, so release year lies on the lineage-to-error path: a child model inherits both the errors of its parent and the training environment of its own release date. Release year is therefore a *mediator* of the lineage path and a *potential confounder* of the observational family–error association (the formal argument is in Appendix B). Any single-number answer that tries to assign shared error "to lineage" or "to era" conflates the two roles of release year.

We address the problem with a variance-decomposition instrument rather than with a causal attribution. Treating the open-model population as a designed experiment, we partition the observed pattern of shared errors into a lineage component σ_L^2, an era component σ_E^2, and a model-specific remainder σ_U^2, estimated with a crossed random-effects (restricted-maximum-likelihood) model. Because the partition is defined only where lineage and era actually vary together—the *connected subset* of the population where the design is crossed rather than nested—the analysis is explicitly scoped to that subset, and an identifiability gate decides whether the decomposition is runnable at all [2], [3].

The central methodological claim of this paper is the ordering of decisions. Identifiability is verified before any trait is measured; the estimator is validated in simulation before any real data are analyzed; and the study population is selected by a pre-analysis design procedure that never observes trait values, before any empirical inference. This ordering is not a matter of tidiness. Each of the three stages can fail, and each failure is one that a conventional analysis would absorb into its results rather than report: a nested design returns a plausible split of an aliased sum; an unvalidated estimator returns a number whose bias is unknown at the sample size in hand; a population chosen after seeing traits returns an effect whose selection is part of the effect. Running the three as gates converts each of these from an unreported assumption into a reported verdict.

The contributions of this paper are as follows.

1) We formulate lineage-versus-era attribution as a crossed variance decomposition on the documented release record, and separate an observational estimand θ_P from a mechanistic estimand θ_M that holds era exactly fixed—two quantities that prior work conflates or reports as one (Section IV).
2) We give sufficient conditions for the partition to be identifiable (Definition 1, Proposition 1), and show that the nested case is a distinct estimand rather than a graceful degeneration of the crossed one (Remark 1).
3) We validate a direct REML estimator against method-of-moments and library implementations, and show it recovers known ground truth to $\leq$ 2.5pp under balanced occupancy and ≤ 5.3pp under the real one, with 95–100% interval coverage (Section VI-B).
4) We specify a detectable-failure battery under which a mis-specified nested design *must* fail visibly, and show three independent detectors flag the aliasing in 100% of repetitions with zero silent coverage (Appendix C).
5) We give an outcome-independent study-population design procedure (Algorithm 2) that selects the smallest identifiable, recoverable population without reading a trait value—22 of 47 models at roughly one-third the measurement cost—and show the structural minimum of 21 is insufficient (Section VI-C).
6) We pre-register the measurement and decision protocol the instrument gates, and release the population, design, optimizer, validator, and tests (Section V-E).

Section II states the research questions; Section III situates the work; Section IV defines the estimands and identifiability conditions; Section V describes the three stages of the protocol (population audit, simulation validation, pre-analysis population design) and specifies the pre-registered measurement protocol the instrument gates; Section VI reports the validated behaviour of the instrument; Sections VII and VIII discuss implications and threats to validity.

II. PROBLEM FORMULATION AND RESEARCH QUESTIONS

A. PRIMARY RESEARCH QUESTION

The main research question (RQ0) asks whether the observed error-correlation structure across public open-weight language models can be separated into a lineage component and an era component; that is, whether the partition $\sigma_L^2 + \sigma_E^2 + \sigma_U^2$ is estimable and non-degenerate on the connected subset. A positive answer requires both: (a) the estimator survives simulation validation under the realistic population occupancy, and (b) on real data, the lineage and era components are each estimated with non-degenerate intervals.

B. REFUTABLE SUB-QUESTIONS AND THEIR FALSIFICATION CONDITIONS

Table 1 lists the sub-questions, their operating population, and their refutation conditions. RQ3 is the primary observational estimand (lineage conditional on era); RQ4 is the mechanistic estimand (lineage with era held fixed), available only on the small co-released and staggered-fine-tune subsets and reported separately. RQ2 is the stop-rule question: a misspecified generative model must fail *detectably* rather than silently, or the program stops.

C. THE TWO-ESTIMAND RULE: OBSERVATIONAL AND MECHANISTIC ATTRIBUTION

RQ3 and RQ4 answer different questions and are never merged. Release year is a mediator of the lineage-to-error path and a potential confounder of the observational family–error association, so the observational primary adjusts for era grouping but cannot hold era fixed; the mechanistic secondary is the only estimand that isolates lineage with era held exactly fixed, and it exists only where co-released family cohorts and verified cross-generation fine-tune chains provide the contrast. We report the two estimands as θ_P (observational) and θ_M (mechanistic), separately, throughout.

III. RELATED WORK AND POSITIONING

Table 2 positions this work against its closest neighbors; this section expands on the three nearest ones.

A. CROSS-SECTIONAL AGREEMENT AND ENSEMBLE CO-FAILURE

Kim *et al.* measure cross-sectional agreement between language models—how often two models agree when both are wrong—across more than 350 open and hosted models on two leaderboards, finding roughly 60% agreement among co-erring models and agreement that persists across distinct architectures and providers [1]. They are explicitly cross-sectional and leave causal and temporal attribution open; they measure a correlation, not a variance partition. On the failure side, Chen shows that pairwise error correlation substantially *underestimates* the co-failure ceiling of an ensemble—the probability that every member fails a question—across 67 frontier models and three combination schemes (routing, voting, mixture-of-agents), by roughly a factor of two [35]. The operative risk is therefore the joint all-wrong rate, whose governing inputs are exactly the lineage and era shares we estimate: our partition quantifies the two components whose sum bounds that ceiling on the connected population, so the two results are complementary rather than competing.

B. VARIANCE COMPONENTS AND GENERALIZABILITY THEORY

The statistical machinery this paper applies is old and well understood; what is new is the object it is applied to. Estimating the variance attributable to each of several crossed grouping factors is the classical variance-components problem, developed for unbalanced designs by Henderson [5] and placed on a likelihood footing by Patterson and Thompson [3] and Harville [2], whose restricted likelihood removes the downward bias that maximum likelihood incurs by ignoring the degrees of freedom spent on fixed effects. Searle, Casella, and McCulloch [6] and Rao and Kleffe [7] give the general theory; Laird and Ware [9] and McCulloch and Searle [8] extend it to the mixed-model and generalized settings, and Pinheiro and Bates [10] and Bates *et al.* [11] supply the computational treatment that most applied work now relies on.

Two aspects of this literature bear directly on our design. The first is small-sample inference. With only six families, the family variance is estimated from five degrees of freedom, and the asymptotic intervals that mixed model software reports by default are known to be optimistic in exactly this regime; Satterthwaite [12] and Kenward and Roger [13] give the standard corrections, and Self and Liang [14] and Stram and Lee [15] characterize the non-standard null distribution that arises when a variance component is tested at the boundary of the parameter space. This is why we report interval coverage from simulation at the actual design size rather than quoting nominal coverage, and why the six-family limit is carried as a documented bound in Section VIII rather than treated as resolved.

The second is that decomposing measurement variance into facets is itself a mature research programme. Generalizability theory [4], [16] was built precisely to ask what fraction of observed variation in a measurement is attributable to persons, items, raters, or occasions, and to use that decomposition to design a measurement procedure before running it. Our contribution is closer in spirit to G-theory than to the LLM evaluation literature: the population of models plays the role of persons, family and release quarter play the role of crossed facets, and the pre-analysis population design of Section V-C is a decision study in G-theory's sense. The multilevel-modelling texts [17], [18] make the same crossed-versus-nested distinction that governs Proposition 1, and Demidenko [19] treats the identifiability conditions directly. What none of this literature supplies is the object: no prior work poses model lineage and release era as the crossed facets of a model population's error variance.

C. VARIANCE DECOMPOSITION IN EVALUATION AND MEASUREMENT PIPELINES

The closest statistical neighbor is the total-evaluation-error (TEE) line [4], [31]. Messing decomposes the uncertainty of LLM evaluation pipelines—judge-model choice, temperature, and prompt phrasing—into crossed random effects via generalizability theory, and shows that naive standard errors are 40–60% smaller than the TEE-corrected ones and that naive confidence intervals under-cover as n grows [31]. The estimator class is the same (crossed random effects, REML), but the grouping factors are properties of the *measurement pipeline*, not of the models; identifiability of a model-population partition is never the question. The two objects are composable: our trait-level decomposition sits on top of a pipeline whose measurement variance TEE-style designs quantify. Classical generalizability theory applies to facets of a measurement procedure rather than to a population of models whose identifiability must be established first [4].

TABLE 1. Research questions, operating data, refutation conditions, and execution status. Each question is paired with the observation that would refute it; the status column records which stages are reported in this paper and which are deferred to the pre-registered measurement protocol of Section V-E.

RQ	Question	Operates on	Refutation condition	Status
RQ1	Does the crossed random-effects estimator recover known ground truth under a balanced-crossed (D1) and a realistic-occupancy (D2) regime?	Simulated data	Bias or collapse under D2	Reported (Sec. VI-B)
RQ2	Under a nested (mis-specified) design, does the estimator fail detectably?	Simulated data	Silent bias	Reported (Sec. VI-B)
RQ3	What share of error covariance is attributable to lineage conditional on era?	Connected subset	$\sigma_L^2 \approx 0$	Pre-registered
RQ4	Holding era exactly fixed, what is the structural contribution of lineage?	Co-released cohorts + fine-tune chains	Reported separately; never merged with RQ3	Pre-registered
RQ5	After lineage adjustment, does the era component converge across release quarters?	Connected subset	Flat trend	Pre-registered
RQ6	Is cross-family diversification a credible mitigation for correlated failure?	Decision layer	Era dominates	Pre-registered

D. LINEAGE RECONSTRUCTION AND MODEL PROVENANCE

Phylogeny-reconstruction work treats lineage as a tree to be recovered from output similarity: PhyloLM builds phylogenetic trees across open and closed models and shows that tree distance predicts benchmark performance [27]. We instead take the documented release record as given and treat lineage as a variance component of a designed population. Tracing the Roots reconstructs the evolution of *datasets* in post-training—83 seed datasets giving rise to 430 datasets and 971 inheritance edges—and shows that benchmark contamination propagates along lineage paths [32]. Dataset ancestry is a channel into our era component (shared training data is a shared environment), and its finding that contamination travels along lineage is exactly the mechanism that motivates the era channel, not a competing estimator of model error covariance. Section IV-E derives the exact mapping from the partition to the co-failure ceiling and the marginal value of a cross-family swap.

E. INSTRUMENT-LEVEL BIAS AND THE ALGORITHMIC MONOCULTURE LITERATURE

Preference leakage shows that LLM-as-judge ratings are biased toward generators that are the same model as, inherit from, or share a family with the judge [33]. This is a downstream *measurement-instrument* consequence of correlated error rather than a trait-level decomposition; it strengthens the case for measuring error structure directly instead of through judge-mediated leaderboards. The monoculture literature documents the systemic risk of deploying many similar models but treats correlated error as a given input to a welfare argument rather than as an object to be decomposed [28], [29]; Jo *et al.* show that monoculture estimates are null-model-dependent and argue that an estimand must be defined independently of a baseline, which is why we model shared random effects directly rather than relying on an independence null [29]. Recent work on behavioral entanglement measures how much the behavior of one model can be predicted from another across six open-weight families, but stops at an independence audit and verifier reweighting rather than a trait-level variance partition [30].

The economic and welfare framing of monoculture predates the language-model case. Kleinberg and Raghavan [25] show that convergence by many decision-makers on a single algorithm can reduce aggregate decision quality even when that algorithm is individually the most accurate, and Bommasani *et al.* [26] formalize outcome homogenization—the extent to which the same individuals experience the same outcome across independent deployments—and measure it for systems that share components. Both take correlated behaviour as the premise from which a welfare conclusion follows. Our question is upstream of theirs: given that correlated failure exists, what generates it, and which of its sources a portfolio designer can act on.

F. ENSEMBLE DIVERSITY AND THE ACCURACY–INDEPENDENCE TRADE-OFF

That ensemble benefit depends on error independence rather than on member accuracy is one of the oldest results in machine learning. Dietterich [21] sets out the conditions under which combination helps, Wolpert [22] and Breiman [23] exploit deliberate decorrelation of members, and Kuncheva and Whitaker [24] evaluate ten diversity measures and find their relationship to ensemble accuracy weaker and less consistent than the folk theory assumes—a caution directly relevant here, since it means a diversity statistic is not interchangeable with the variance structure that actually governs joint failure. This literature almost universally assumes members are trained independently by construction, which is exactly the assumption that fails for a portfolio of public language models: members are drawn from a population with shared ancestry and a shared release environment, neither of which the deploying organization controls. The lineage/era partition is a way of measuring how badly that assumption fails, and along which axis.

G. POSITIONING: WHAT REMAINS UNMEASURED

Across these five literatures the pattern is consistent. The agreement and co-failure work measures that models err together but reports a correlation, not a decomposition, and is cross-sectional by construction. The variance-components and generalizability literature supplies exactly the decomposition required, but has never been pointed at a model population, and so has never had to confront the fact that the two candidate facets are confounded by release chronology. The pipeline-variance work decomposes a different object—the measurement apparatus rather than the models. The lineage-reconstruction work recovers ancestry structure but stops before attributing error variance to it. The monoculture and ensemble literatures consume correlated error as an input to a welfare or combination argument. Table 2 states the gap for each.

What follows from this is not merely that the partition is unmeasured. It is that measuring it requires settling a question none of these literatures has had to ask: whether the lineage and era components are separately identifiable on the population actually available, given that ancestry and release date move together. That question has a determinate answer, it can be settled before any measurement, and it can fail. The rest of this paper is about settling it.

No verified prior work estimates $\sigma_L^2 + \sigma_E^2 + \sigma_U^2$ for model error traits, on the provably connected subset, with the estimator's validity established in simulation first. Every prior approach either documents correlation without partitioning it, decomposes the variance of a different object, or treats correlated error as a given input to a downstream argument.

IV. FORMAL MODEL, ESTIMANDS, AND IDENTIFIABILITY CONDITIONS

A. POPULATION MODEL AND NOTATION

Let the population consist of N open-weight models indexed by $i \in \{1, \ldots, N\}$. Each model is released by a family $f(i) \in \{1, \ldots, 6\}$ (a verified design grouping, e.g., Llama, Mistral, Qwen, Phi, Gemma, DeepSeek) in a release quarter $e(i) \in \{1, \ldots, 14\}$ spanning 2023Q1–2026Q2. Each model receives a continuous per-model trait Y_i (Section V), measured on a fixed common item set. We model the trait as

$$Y_i = \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \tag{1}$$

where μ is the grand mean, $\alpha_f \sim \mathcal{N}(0, \sigma_L^2)$ are independent family (lineage) effects, $\beta_e \sim \mathcal{N}(0, \sigma_E^2)$ are independent era effects, and $u_i \sim \mathcal{N}(0, \sigma_U^2)$ are independent model-specific residuals. Table 3 collects the notation.

In vector form,

$$\mathbf{y} = \mu\mathbf{1} + \mathbf{C}\boldsymbol{\gamma} + \mathbf{u}, \qquad \boldsymbol{\gamma} \sim \mathcal{N}\big(\mathbf{0}, \mathrm{diag}(\sigma_L^2, \sigma_E^2)\big), \tag{2}$$

The causal structure that motivates the two-estimand rule is summarized in Fig. 1: family membership influences the error trait directly and through the release date, and release date is a mediator of the lineage path and a potential confounder of the observational family–error association. Because the mediational path cannot be separated from the lineage channel without holding era fixed, a single causal attribution is impossible by construction.

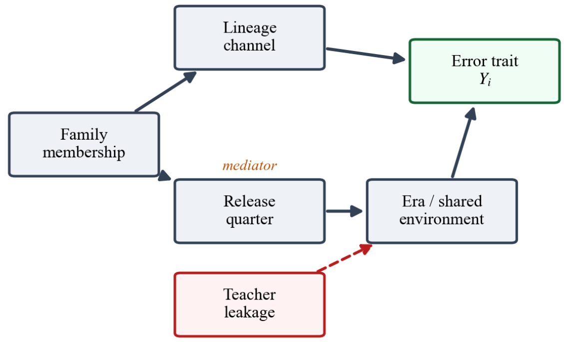

FIGURE 1. Causal structure of the error trait. Family membership and era both influence the error trait; release date is a mediator of the lineage path and a potential confounder of the observational family–error association, motivating the separate observational (θ_P) and mechanistic (θ_M) estimands. Teacher leakage enters the era (shared environment) channel. Appendix B gives the formal identification argument.

where $\mathbf{C}$ is the $N \times 12$ design matrix of family and era indicators with u_i as in (1). The marginal covariance is

$$\mathbf{V}(\theta) = \sigma_U^2\mathbf{I} + \mathbf{C}\,\mathrm{diag}(\sigma_L^2, \sigma_E^2)\mathbf{C}^\top, \tag{3}$$

with $\theta = (\sigma_L^2, \sigma_E^2, \sigma_U^2)$.

B. RESTRICTED-MAXIMUM-LIKELIHOOD ESTIMATION OF THE VARIANCE COMPONENTS

We estimate θ by restricted maximum likelihood (REML), maximizing the log-restricted-likelihood [2], [3]

$$\begin{aligned}\ell_R(\theta) = & -\tfrac{1}{2}\log\det\mathbf{V} - \tfrac{1}{2}\log\det(\mathbf{1}^\top\mathbf{V}^{-1}\mathbf{1}) \\ & -\tfrac{1}{2}(\mathbf{y}-\mu\mathbf{1})^\top\mathbf{V}^{-1}(\mathbf{y}-\mu\mathbf{1}),\end{aligned} \tag{4}$$

maximized directly with a numerical optimizer. Because $\mathbf{C}$ is low-rank ($F + E = 20$ columns), the Woodbury identity

$$\mathbf{V}^{-1} = \sigma_U^{-2}\Big(\mathbf{I} - \mathbf{C}\big(\sigma_U^2\mathbf{I} + \mathrm{diag}(\sigma_L^2, \sigma_E^2)\mathbf{C}^\top\mathbf{C}\big)^{-1}\mathbf{C}^\top\Big) \tag{5}$$

is used throughout, and is stated as Algorithm 1. It reduces each likelihood evaluation to linear algebra on 20×20 matrices, making repeated simulation feasible. Share confidence intervals use a Monte-Carlo delta method on the log-variances with a numerically differentiated Hessian, PSD-clipped for sparse designs.

We verified this direct implementation against the standard reference implementations. On balanced data the REML estimates reproduce two-way ANOVA method-of-moments estimates exactly (verified at 12×12, 6×14, and 8×10 designs). In contrast, the widely used ‘statsmodels‘ crossed-variance formulation does *not* maximize the REML objective (Table 4): on a balanced 12×12 design it returns a family/era split away from the optimum (REML objective 66.674 vs. 65.994 for the direct optimizer), and on the realistic D2 occupancy it understates the family share by roughly $5\times$ more than the documented six-family small-sample limit. This is an upstream finding worth reporting; all results here use the

TABLE 2. Differentiation against closest related work. The final column states what each line of work leaves unmeasured; the union of those gaps is the object of this paper.

Work	Object	What it does	What it leaves open
Kim *et al.*, ICML 2025	agreement rate	measures when models co-err; cross-sectional	no variance partition; no temporal structure
Messing, TEE (2026)	pipeline facets	crossed random effects on judge/temperature/prompt; TEE-corrected CIs	facets are pipeline, not model population; no identifiability gate
Li *et al.*, Tracing the Roots (ACL 2026)	dataset lineage graphs	reconstructs post-training data ancestry; contamination along paths	dataset lineage is a channel into the era component, not an error-trait partition
Li *et al.*, Preference Leakage (ICLR 2026)	judge-model bias	judges favor related generators (same/inherit/family)	instrument-level bias, downstream of correlated error
PhyloLM / lineage reconstruction	phylogeny	reconstructs ancestry trees	no variance decomposition
Kuai *et al.*, 2026	behavioral entanglement	independence audit + verifier reweighting	entanglement indices, not a trait partition
Monoculture literature	deployed portfolio	welfare argument given correlated error	correlated error is an input, not the estimand
This work	error traits on connected subset	identifiability-gated $\sigma_L^2/\sigma_E^2/\sigma_U^2$ partition	—

TABLE 3. Notation.

Symbol	Meaning
(a) Design and population	
N, i	size of the open-model population (47); model index
$f(i), e(i)$	family and release quarter (era) of model i
F, E	number of families (6) and release quarters (14)
$\mathbf{C}$	$N \times (F+E)$ design matrix of family and era indicators
VIF	variance inflation factor of the design columns
(b) Model, components, and estimands	
Y_i	continuous per-model error trait, measured on the fixed common item set
μ	grand mean
α_f, β_e, u_i	family (lineage), era, and model-specific random effects in (1)
$\sigma_L^2, \sigma_E^2, \sigma_U^2$	lineage, era, and model-unique variance components
θ	variance-component vector $(\sigma_L^2, \sigma_E^2, \sigma_U^2)$
θ_P	observational share estimand (lineage conditional on era grouping)
θ_M	mechanistic share estimand (lineage with era held exactly fixed)
$\mathbf{V}(\theta)$	marginal covariance of $\mathbf{y}$ given (3)
$\ell_R(\theta)$	log-restricted-likelihood (REML objective, (4))

Algorithm 1 REML fit via Woodbury with log-variance reparameterization

input trait $\mathbf{y} \in \mathbb{R}^N$; design $\mathbf{C} \in \mathbb{R}^{N\times(F+E)}$; tolerance ϵ
$\psi \leftarrow (\log \hat\sigma_L^2, \log \hat\sigma_E^2, \log \hat\sigma_U^2)$ from a method-of-moments start ▷ *unconstrained parameterization keeps the optimizer feasible*
repeat
  $(\sigma_L^2, \sigma_E^2, \sigma_U^2) \leftarrow \exp(\psi)$
  $\mathbf{G} \leftarrow \mathrm{diag}(\sigma_L^2 \mathbf{I}_F, \sigma_E^2 \mathbf{I}_E)$
  $\mathbf{M} \leftarrow \sigma_U^2 \mathbf{I} + \mathbf{G}\,\mathbf{C}^\top\mathbf{C}$ ▷ $(F{+}E) \times (F{+}E) = 20 \times 20$; *$\mathbf{C}^\top\mathbf{C}$ is built once*
  $\mathbf{V}^{-1} \leftarrow \sigma_U^{-2}(\mathbf{I} - \mathbf{C}\mathbf{M}^{-1}\mathbf{C}^\top)$ ▷ *Woodbury,* (5)
  $\log\det \mathbf{V} \leftarrow (N - F - E)\log\sigma_U^2 + \log\det\mathbf{M}$ ▷ *matrix determinant lemma*
  evaluate $\ell_R(\psi)$ by (4); take a quasi-Newton step in ψ
until $\|\nabla \ell_R\| < \epsilon$
$\hat\theta \leftarrow \exp(\psi)$; clip to the PSD cone ▷ *sparse designs can drive a component to the boundary*
$\mathbf{H} \leftarrow$ numerically differentiated Hessian of ℓ_R at ψ
draw $\psi^{(b)} \sim \mathcal{N}(\psi, -\mathbf{H}^{-1})$, $b = 1, \ldots, B$; map through the share transform
return $\hat\theta$, $\hat\theta_P$, and the Monte-Carlo share intervals

direct maximizer. The exact comparison—optimizer settings, design, and the two objective values—is reproduced in the released artifact (src/lineage_era/analysis/reml.py) and in the project decision log (2026-08-03); a fresh reproduction of the family-share gap on D2 (scenario A) is included in Table 4.

C. OBSERVATIONAL AND MECHANISTIC ESTIMANDS

The primary observational estimand is the share vector

$$\theta_P = \left(\frac{\sigma_L^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2}, \frac{\sigma_E^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2}, \frac{\sigma_U^2}{\sigma_L^2 + \sigma_E^2 + \sigma_U^2} \right). \tag{6}$$

The mechanistic estimand θ_M is the lineage share within designs that hold era exactly fixed: co-released family cohorts and the verified cross-generation fine-tune chains (the only documented parent–offspring edges), reported separately from θ_P.

D. SUFFICIENT CONDITIONS FOR IDENTIFIABILITY OF THE PARTITION

The partition (6) is estimable only if the design is crossed in a sense we now make precise.

Definition 1 (Connected crossed design)**.** A design (f, e) on a model population $\mathcal{M}$ is *crossed* if every family spans at least two release quarters and at least two quarters contain at least two families. It is *connected* if the bipartite family–era incidence graph—one vertex per family, one per quarter, an edge whenever some model occupies that cell—is connected.

Connectivity is what licenses comparison across the whole population: in a disconnected design the components carry separate, non-comparable grand means, and a partition estimated across them is a partition of an artifact. Crossing is what separates the two variance components. The following makes the second claim exact; the proof is in Appendix D.

TABLE 4. Estimator comparison: the direct maximizer vs. the 'statsmodels' crossed-variance path. Balanced design: *F*=*E*=12, two models per cell (288 models), fixed simulated dataset as in the decision log (2026-08-03); the family-share bias column is measured on the realistic D2 occupancy (scenario A), reproduced 2026-08-06.

Method	Fam. σ^2	Era σ^2	Uniq. σ^2	REML obj. ($-2\ell_R$)	Fam.-share bias on D2 (A)
Two-way ANOVA (method of moments)	0.399	0.313	0.656	65.994	—
Direct REML (this work, Woodbury)	0.399	0.313	0.656	65.994	−5.3pp (300 reps)
'statsmodels' MixedLM (*vc_formula*)	0.609	0.476	0.656	66.674	−28.0pp (40 reps)

The direct maximizer and the method-of-moments reference agree to three decimals on the balanced design and reach the same restricted log-likelihood; the `statsmodels` crossed-variance path stops 0.68 higher in $-2\ell_R$, i.e. short of the optimum, and inflates both variance components.

Proposition 1 (Crossing implies identifiable rank)**.** *Let* $\mathbf{C} = [\mathbf{Z}_F\ \mathbf{Z}_E]$ *be the* $N \times (F+E)$ *incidence matrix of a connected crossed design in the sense of Definition 1. Then* $\mathbf{C}$ *has rank* $F+E-1$*, the deficiency being exactly the intercept direction* $\mathbf{1}$*, and* $\sigma_L^2, \sigma_E^2, \sigma_U^2$ *are separately estimable from* $\mathbf{V}(\theta)$*.*

Operationally we therefore require: (i) the design is crossed and connected; (ii) $\operatorname{rank}(\mathbf{C}) = F+E-1$; and (iii) the variance-inflation factors of the design columns are bounded (VIF ≤ 10), which is a numerical stability requirement rather than an identifiability one—a design can satisfy (i) and (ii) and still be so nearly aliased that the components are not recoverable at realistic sample sizes. This is precisely the gap that the simulation stage of Section V exists to close.

Remark 1 (The nested case is a different estimand, not a degenerate one)**.** When each family is confined to a single era, $\mathbf{Z}_F$ lies in the column span of $\mathbf{Z}_E$ and σ_L^2 and σ_E^2 are perfectly aliased: no estimator can separate them, and their sum—not either one—is what the data identify. It is tempting to read this as the crossed model degenerating gracefully, but it is not. A likelihood maximizer handed a nested design will return a finite, plausible-looking split of that sum, chosen by whatever regularization or starting point the implementation happens to carry. The quantity reported is then well defined only relative to the software. This is why the design battery of Section V treats the nested regime as a *must-fail* case: the requirement is not that the estimator degrade, but that it be detected doing so.

E. CO-FAILURE CEILING AND THE DIVERSIFICATION COUNTERFACTUAL

The partition (6) is descriptive; this subsection gives it decision content. For a pool S of k models, define the *co-failure ceiling* $\beta(S)$ as the probability that every member of S is wrong on a fresh question—the joint all-wrong rate that Chen identifies as the operative risk for ensembles and routing [35]. Under the liability model of (1) with the item response $\Pr(\text{wrong} \mid i) = \sigma(-l_m - \delta_i)$ and a shared item difficulty $\delta_i \sim \mathcal{N}(0, \sigma_\delta^2)$,

$$\beta(S) = \mathbb{E}\Big[\prod_{m \in S} \sigma(-l_m - \delta_i)\Big], \tag{7}$$

where the expectation is over the trait effects, the item difficulty, and the item-level noise. Using the logistic–normal link $\sigma(x) \approx \Phi(x/\kappa)$ with $\kappa = 1.6$, centering the trait at its mean μ_l, and writing $W_m = l_m - \mu_l + \delta_i + \eta_m$ with $\eta_m \sim \mathcal{N}(0, \kappa^2)$ i.i.d., the all-wrong event $\{l_m + \delta_i + \eta_m < 0 \text{ for all } m\}$ is $\{W_m < -\mu_l \text{ for all } m\}$, so

$$\beta(S) = \Phi_k(-\mu_l \mathbf{1}; \mathbf{0}, \Sigma), \qquad \Sigma_{mm} = \sigma_L^2 + \sigma_E^2 + \sigma_U^2 + \sigma_\delta^2 + \kappa^2, \tag{8}$$

with off-diagonal $\Sigma_{mm'} = \sigma_L^2 \mathbf{1}[f(m){=}f(m')] + \sigma_E^2 \mathbf{1}[e(m){=}e(m')] + \sigma_\delta^2$: the covariance the partition estimates, plus the item-level and link terms common to any pool.

Proposition 2 (The ceiling is bounded by the between-model share)**.** *$\beta(S)$ is strictly increasing in every off-diagonal entry of Σ, hence in σ_L^2 and in σ_E^2. Consequently $\sigma_L^2 + \sigma_E^2$ pins the co-failure ceiling: no pool drawn from the partition can lower the all-wrong rate below the level implied by $\sigma_L^2 + \sigma_E^2$, whatever its composition.*

The quantity relevant to RQ6 is the marginal value of cross-family diversification. Swapping one member m of S for a model m^* from an unrepresented family changes only the covariances touching the replaced member:

$$\Sigma_{m'm^*} - \Sigma_{m'm} = -\sigma_L^2 + \sigma_E^2 \big(\mathbf{1}[e(m^*){=}e(m')] - \mathbf{1}[e(m){=}e(m')]\big). \tag{9}$$

Proposition 3 (Swap value is governed by the lineage–era ratio)**.** *Let $\Delta\beta = \beta(S) - \beta(S')$ for a same-size swap into an unrepresented family. Then $\Delta\beta \geq 0$; it is non-decreasing in σ_L^2 and non-increasing in σ_E^2, and it grows when the replacement's era matches no pool member's era, because the σ_E^2 term of (9) then vanishes. The marginal value of diversification is thus governed by $\sigma_L^2/(\sigma_L^2 + \sigma_E^2)$, not by the unique share σ_U^2: model-specific noise is diversifiable within a family for free, while era-common error is not removed by a family swap.*

Propositions 2 and 3 are closed forms of the model in (1); we verify them against the exact generative process in simulation (`test_cofailure.py`): the analytic orthant (8) tracks the Monte-Carlo all-wrong rate of (7) to within 0.03 across the lineage-dominant, era-dominant, and balanced scenarios, and the swap ordering of Proposition 3 holds in each. The empirical value of β for the measured population is a stage-4 deliverable and is reported only once the partition is estimated on real data.

V. METHODOLOGY: A THREE-GATE PROTOCOL FOR IDENTIFIABILITY-PRESERVING MEASUREMENT

The protocol proceeds in three stages, each with an exit gate. The ordering is the contribution: identifiability before results, simulation before real data, and study-population design before empirical inference. Fig. 2 shows the protocol spine; stages 1–3 are complete and reported here, and stage 4 is the pre-registered measurement pass whose outcomes are reported against the pre-registered decision rule.

A. STUDY POPULATION CONSTRUCTION

We assembled a verified population of $N = 47$ open-weight general models spanning six families and fourteen release quarters (2023Q1–2026Q2). The family × quarter contingency table was built from Hugging Face API metadata and technical reports, with documented divergences between the public release date and HF timestamps corrected by hand. Two membership rules are central. First, *all claims are scoped to the connected subset*—the maximal subset on which lineage and era vary jointly—so the partition (6) is defined. Second, hosted or closed-weight models are excluded. The resulting design is crossed but unbalanced: 11 of 14 quarters contain at least two families and no family is confined to a single era (Fig. 3).

B. SIMULATION VALIDATION OF THE ESTIMATOR

Before any real data are analyzed, the estimator is validated on simulated data with known ground truth under three regimes [2]:

- *D1, balanced crossed reference:* 30 families × 14 eras × 2 models per cell; the large family count isolates estimator calibration from the small-sample limit of the real design.
- *D2, realistic occupancy:* The occupancy matrix copied from the study population table (6 families × 14 quarters, 47 models, sparse cells). Bias or collapse here is a gate failure: the real-data claim never proceeds.
- *D3, nested (must fail):* Each family confined to a single era, so σ_L^2 and σ_E^2 are perfectly aliased. Three independent detectors (BLUP family-vs-era collinearity, SE/estimate inflation, and profile-likelihood flatness) must flag the aliasing.

A liability decision was resolved in simulation before the trait path was fixed: on item-level binary outcomes at the real occupancy, both a linear-probability model and a binomial GLMM under-estimate the era component and drive it to the boundary in 20–60% of repetitions, whereas continuous per-model traits recover it with 98–100% coverage. Phase 2 therefore uses per-model continuous traits with LPM–REML. A family × era interaction term was also simulated and documented as non-identified at the sparse occupancy (SE/estimate ratios $\approx 10^4$); the interaction is excluded by design rather than absorbed.

TABLE 5. G3 optimization inputs.

Input	Used
Family × quarter occupancy	✓
Lineage graph (verified edges, chain)	✓
Identifiability constraints (rank, VIF, span, crossing)	✓
Cost (public > gated, est. GPU minutes)	✓
Trait values / accuracy	×
Error similarity	×
Evaluation outputs	×

C. PRE-ANALYSIS STUDY-POPULATION DESIGN

The empirical campaign measures a fresh trait on all 47 models. Because a full run is expensive, we pre-registered a gate that selects the study population *before any GPU spend*: the smallest connected-subset population whose family × quarter design is structurally identifiable *and* whose era recovery clears the strict Phase 1 bar in simulation. The key property of this stage is that it is *outcome-independent*: the optimizer's inputs are occupancy (family × quarter), the lineage graph, identifiability constraints, and cost—never trait values, accuracies, or evaluation outputs (Table 5).

The procedure is fully pre-registered. It minimizes model count subject to hard constraints (all six families present with at least two quarters of span; every quarter 2023Q1–2026Q2 retains at least one model; the in-subset verified-edge endpoints and the documented chain are retained so the mechanistic estimand survives; the induced design passes the structural checks), then tie-breaks on cost. "Valid," not merely "identifiable," requires that the candidate design recover known era shares in a fixed-design DGP battery (scenarios A and B) under the strict bar (absolute era-share bias $\leq$ 5pp, CI coverage $\geq 90\%$, convergence $\geq 90\%$). Because the per-rep share bias is heavy-tailed at this occupancy (standard deviation 22–26pp), a candidate that clears the bar at 300 repetitions must also clear it with a $\geq$ 1pp margin at a 1000-repetition confirmation; knife-edge designs sitting exactly on the bar are rejected. Kept models are assigned an auditable reason by single-model ablation—removing the model and asking which design constraint fails—so the population file documents *why* each model is required (era-window, structural identifiability, or statistical recoverability).

Algorithm 2 states the procedure. Its defining property is the assertion on line 2: the trait vector is never an input, so no branch of the search can depend on an outcome. This is what distinguishes the procedure from a power analysis run after a pilot measurement, and it is why the resulting population can be published before the measurement without creating a researcher degree of freedom.

D. TRAIT MEASUREMENT AND DECOMPOSITION

On the selected population, the pipeline is: (i) a fresh five-shot MMLU pass [34] on the common fixed item set (the only compliant source of per-question data; a public reuse audit found no reusable per-question records); (ii) intake validation that aborts on any contract violation of the manifest; (iii) assembly of the continuous per-model trait; (iv) the identifia-

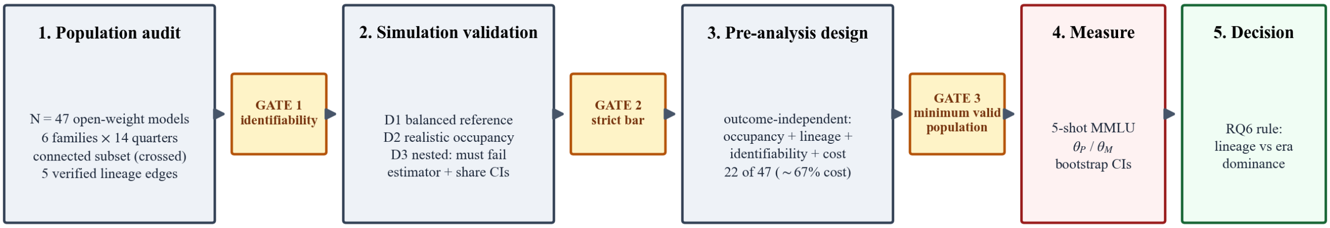

FIGURE 2. Protocol architecture. Three completed stages (population audit, simulation validation, pre-analysis study-population design), each behind an identifiability or recoverability gate, followed by the measurement and decomposition stage the instrument gates and the decision layer. Stages 1–3 constitute the instrument and are established in this paper; stage 4 is specified in full by the pre-registered protocol of Section V-E. The G3 stage is outcome-independent: it never observes trait values.

Algorithm 2 Outcome-independent study-population design (G3)

input model set $\mathcal{M}$ ($|\mathcal{M}| = 47$); occupancy O: family × quarter; verified lineage edges L; per-model cost c; bars τ_{bias}=4.0pp, τ_{cov}=90%, margin δ=1pp
assert trait values $\mathbf{y}$ are not readable in this scope ▷ *outcome-independence invariant*
$\mathcal{S} \leftarrow \{\, S \subseteq \mathcal{M} : S$ satisfies the hard constraints $\}$, namely
  all F families present in S
  every quarter of the era window retains ≥ 1 model
  both endpoints of every edge in L retained
  the documented within-family chain retained
for $S \in \mathcal{S}$ sorted by $\sum_{i \in S} c_i$ ascending **do**
  build $\mathbf{C}_S$; **if** $\operatorname{rank}(\mathbf{C}_S) < F + E - 1$ **then continue**
  if $\max \operatorname{VIF}(\mathbf{C}_S) > 10$ **then continue** ▷ *structural identifiability*
  if S has an era with < 2 families or a family with < 2 eras **then continue**
  $(b, \kappa) \leftarrow \mathit{Screen}(S, R{=}300)$ ▷ *simulated traits only*
  if $|b| > \tau_{\text{bias}}$ **or** $\kappa < \tau_{\text{cov}}$ **then continue**
  $(b, \kappa) \leftarrow \mathit{Confirm}(S, R{=}1000)$
  if $\tau_{\text{bias}} - |b| < \delta$ **then continue** ▷ *reject knife-edge designs*
  if $\mathit{Robust}(S, R{=}2000)$ fails **then continue**
  for $m \in S$ **do** ▷ *auditable reason by single-model ablation*
    reason[m] $\leftarrow$ the first constraint that fails on $S \setminus \{m\}$
  return S, reason
return $\varnothing$ ▷ *no admissible population; the gate blocks measurement*

bility gate on the measured design (hard fail = abort); (v) the θ_P partition with REML (4); (vi) bootstrap confidence intervals combined with a trait-error Monte Carlo; and (vii) sensitivity blocks (leave-one-family, leaked-drop, subject-drop, trait-definition, and a documented sanity cross-check against prior leaderboards that is explicitly not a validation). The era-convergence trend and the θ_M mechanistic tables are reported as table entries, separately from the primary partition.

E. PRE-REGISTERED MEASUREMENT AND DECISION PROTOCOL

The instrument is only as binding as the specification it is committed to before the trait is observed. We therefore fix, in advance, the complete set of outputs the measurement pass will produce, the form each takes, and the rule that maps them onto the RQ6 decision. Nothing in this specification depends on a trait value, and none of it may be revised after the pass is run; Table 6 records it in full.

TABLE 6. Pre-registered measurement outputs and their form. Fixed before the measurement pass; not revisable after it. Every entry is a specification, not a result.

Output	Specification
Error-similarity panel	Pairwise error overlap across the selected models, chance-corrected with ϕ (locked), read against the four-rung null ladder; reported as heatmap, network, and embedding. Supporting, not confirmatory
θ_P partition	REML estimates and bootstrap intervals for $\sigma^2_L, \sigma^2_E, \sigma^2_U$ on the 22-model population; the primary output
Identifiability audit	Rank, VIF, and the three D3 detectors re-run on the measured occupancy; hard failure aborts
Era-convergence entry	Era share by release quarter after lineage adjustment; reported as a table entry, never promoted to a headline figure
θ_M tables	Family contrasts within co-released quarters and per-quarter slopes along the verified fine-tune edges; reported separately from θ_P
Sensitivity battery	Leave-one-family, leaked-drop, subject-drop, trait-definition, and the leaderboard sanity cross-check (not a validation)

Three commitments in that specification deserve emphasis, because each closes a degree of freedom that a post-hoc analysis would otherwise retain. First, the chance-corrected overlap measure is *locked* to ϕ by the pre-registration rule, so the reported co-failure structure cannot be selected from a family of agreement statistics after the overlaps are seen; the null ladder against which it is read—observed → matched-accuracy → item-difficulty → independence—is fixed with it. Second, the identifiability gate is re-run on the *measured* design rather than the planned one: rank, variance-inflation factors, and the three D3 detectors of Appendix C are evaluated on the occupancy that the pass actually returns, and a hard failure aborts the analysis rather than downgrading it to a caveat. Third, the two estimands are reported in separate tables under all circumstances, so that a weak θ_M cannot be quietly absorbed into a stronger θ_P, or the reverse.

The decision rule itself is deliberately asymmetric and is stated in Section VII. Its asymmetry is the reason it must be fixed in advance: a symmetric rule would be robust to the order in which the shares are inspected, whereas this one is not.

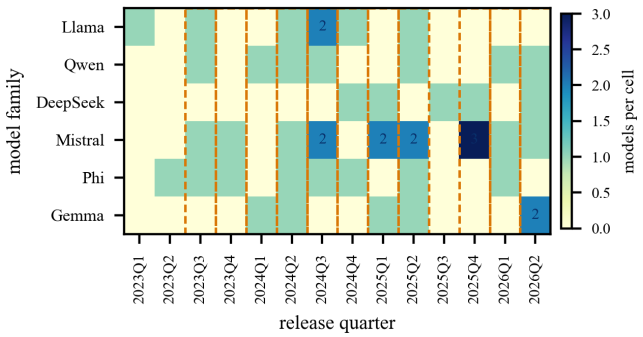

FIGURE 3. Study population construction: family × release-quarter occupancy of the 47-model connected subset. 11 of 14 quarters contain at least two families (highlighted) and no family is confined to a single era; sparse cells carry the small-sample limits reported in Section VIII. Cell shading gives the model count; only cells holding more than one model are annotated, since 39 of the 47 models occupy a cell of their own. The verified cross-generation lineage edges are shown in Fig. 4.

VI. RESULTS

This section reports the three gated stages of the instrument: the structural audit of the model population, the recovery and detectable-failure behaviour of the estimator in simulation, and the population selected by the outcome-independent design procedure. Each is a property of the instrument rather than of any measured trait, and each is established before any trait value is observed—which is precisely what makes the pre-registered measurement protocol of Section V-E binding rather than advisory. The lineage-versus-era shares themselves are the object of that protocol and are deliberately not estimated here.

A. STRUCTURAL AUDIT OF THE 47-MODEL CONNECTED SUBSET

The verified population is 6 families × 14 quarters with $N = 47$ models. The design is crossed (unbalanced/incomplete): 11 of 14 quarters contain at least two families, no family is confined to one era, and 5 cross-generation parent–offspring edges are verified from the metadata (the bulk of true lineage is undocumented and drawn from technical reports). The connected subset—on which all claims are scoped—is the population itself; a nested subpopulation exists and is excluded by the scoping rule. Three caveats are carried into the analysis: the Llama lineage terminates at Llama 4 (post-2025 era variation is carried by the other families); DeepSeek V4 is a ground-up redesign treated as a new independent lineage rather than a descendant; and cross-family teacher leakage (e.g., Phi-4 trained on GPT-4o-generated data) belongs in the era channel as shared environment. Fig. 3 shows the occupancy matrix.

B. ESTIMATOR RECOVERY AND DETECTABLE FAILURE

Table 7 summarizes the validation. Under the balanced reference (D1, 300 repetitions) share bias is $\leq$ 2.5pp with coverage 95–96%. Under the real occupancy (D2) share bias is ≤ 5.3pp with coverage 95–100%; the family-share bias of -5.3pp in the lineage-dominant scenario is the documented small-sample limit of six family levels (df$= 5$) and is reported as a limit, not corrected away. The nested design (D3) is detected in 100% of repetitions by all three independent detectors, with zero silent coverage (Table 8). The liability test resolved the trait path to per-model continuous traits; the family × era interaction is documented as non-identified rather than estimated. The verdict is GO WITH CHANGES, with the changes being the continuous-trait path and the disclosure of the family-share limit.

C. SELECTION OF THE MINIMUM VALID POPULATION UNDER THE G3 GATE

Fig. 5 and Table 9 report the gate. The structural minimum is 21 models (identifiable: full rank, VIF $\leq$ 10), but the 21-model design sits exactly on the strict bar in simulation: its era-share bias at 1000–2000 repetitions is ≈ -5.0pp with a standard error of 0.6–0.8pp—a knife-edge whose verdict flips with the repetition count—so it fails the margin confirmation. The minimum *valid* population is 22 of 47 models, which clears the bar at 300 repetitions (era bias A 2.4pp / B -0.8pp), passes the 1000-repetition margin confirmation (A 2.2pp / B -2.4pp), and is robust at 2000 repetitions (A 1.7pp / B -3.2pp). The full 47-model design passes by the same criterion (A 1.5pp / B -2.2pp at 1000 repetitions), confirming that the reduction is statistically justified: 22 models recover era shares in the same order of magnitude as all 47, at roughly one-third of the estimated single-GPU cost (a 67% reduction). The winner keeps all six families, all fourteen quarters, the verified-edge endpoints, and the chain needed for the mechanistic estimand. The population file assigns each of the 22 kept models an auditable reason by single-model ablation (era-window, structural identifiability, or statistical recoverability) and each dropped model the reason "redundant in-cell replication."

VII. DISCUSSION: OPERATIONAL IMPLICATIONS FOR MODEL PORTFOLIO DESIGN

A. WHY THE PARTITION, AND NOT THE CORRELATION, IS THE DECISION QUANTITY

The quantity practitioners can already measure is the pairwise error correlation between two models, and the quantity they actually need is the probability that an entire portfolio fails the same item. These are not the same, and the gap between them is not small. Chen reports that pairwise correlation substantially *underestimates* the co-failure ceiling—the probability that every member of an ensemble is wrong together—by roughly a factor of two across 67 frontier models and three combination schemes [35]. The reason is structural: pairwise correlation is an average over model pairs and is insensitive to whether the shared component is carried by one group of models or spread across all of them. A portfolio with a modest average correlation concentrated within a single family behaves very differently from one with the same average spread uniformly, and only the second is helped by adding members.

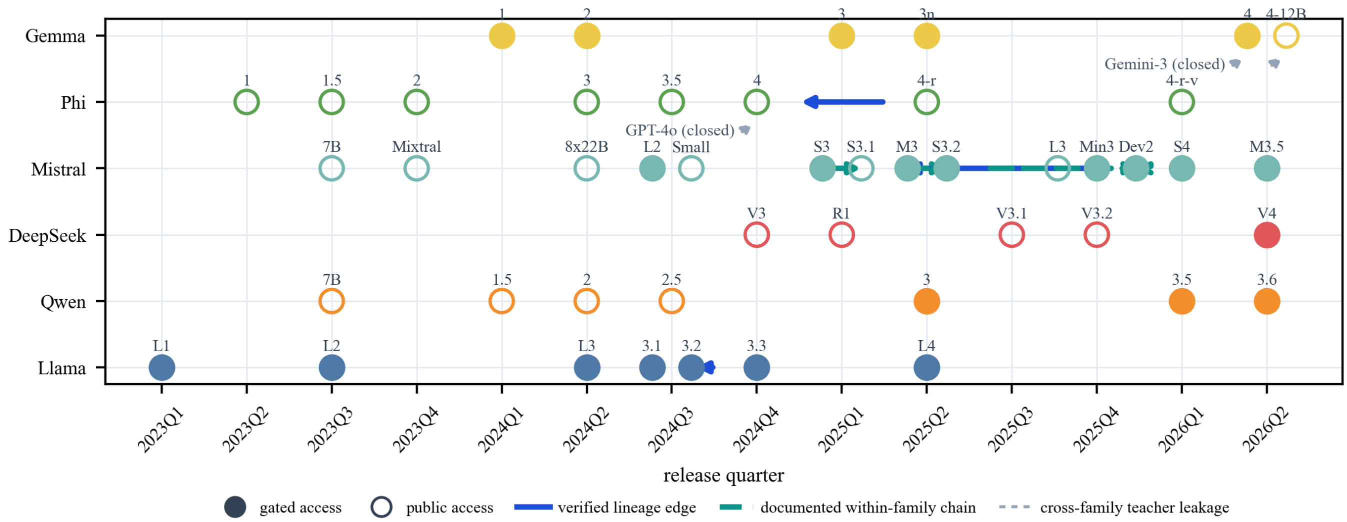

FIGURE 4. Verified lineage on the connected subset. All 47 models on the family × release-quarter grid (filled markers: gated access; open markers: public); models sharing a family–quarter cell are offset within the cell so that each is separately labelled. The five parent–offspring edges verified from the Hugging Face `base_model` field and the documented Mistral-Small chain support the mechanistic estimand θ_M; the primary partition θ_P uses the verified family grouping instead of the sparse edge record. Dashed grey arrows mark the documented cross-family teacher-leakage channels, which belong to the era (shared-environment) channel.

TABLE 7. Simulation validation (Phase 1). Each regime is run to a fixed repetition budget on a recorded seed and carries an explicit gate verdict; D3 is a must-fail regime, so its PASS records that the aliasing was detected, not that the partition was recovered.

Regime	Recovery / detection	Reps	Gate	Verdict
D1 balanced-crossed	share bias $\leq$ 2.5pp, coverage 95–96%	300	PASS	GO
D2 realistic occupancy	share bias $\leq$ 5.3pp, coverage 95–100%	300	PASS[a]	GO (documented)
D3 nested (aliased)	detection 100%, silent coverage 0%	300	PASS	GO
Liability (binary)	era underpowered at real occupancy; continuous recovers	300	n/a	path decided
L×E interaction	non-identified (SE ratio $\approx 10^4$)	300	n/a	documented

[a]The strict bar is defined on the *era*-share bias (mean over converged repetitions; Section V-B); the family-share bias of -5.3pp in the lineage-dominant scenario is the six-family small-sample limit (df$=5$) and is reported as a documented limit in Section VIII, not waived.

TABLE 8. Detectable-failure battery under the nested (mis-specified) design. Each detector is evaluated independently on 300 fresh repetitions per generative scenario on a fixed seed. The gate requires *all three* to flag the aliasing in $\geq$ 90% of repetitions with no repetition showing silent confidence-interval coverage; a nested design that appears to succeed is a broken detector, not a valid estimate.

Detector	Statistic	Threshold	Sc. A	Sc. B
BLUP collinearity	$\lvert\mathrm{corr}(\hat{u}_F, \hat{u}_E)\rvert$	> 0.9	100%	100%
SE inflation	$\widehat{\mathrm{SE}} \geq \lvert\hat{\sigma}\rvert$	ratio ≥ 1	100%	100%
Profile flatness	profile drop over ± 0.4 window	< 1.9207	100%	100%
Joint detection	any detector fires	$\geq 90\%$	**100%**	**100%**
Silent CI coverage	undetected reps covering truth	$= 0$	0	0

Thresholds are defined in Appendix C; 1.9207 is the χ^2_1 95% critical value halved. Non-convergence is tracked as a fourth signal and did not fire independently of the three above.

A variance partition distinguishes these cases directly, because it says *which* grouping carries the shared component. That is why the decision rule below is stated on σ^2_L and σ^2_E rather than on an agreement rate: the shares, not the correlation, are what determine whether adding a model from a new family buys independence or merely buys another draw from the same environment.

TABLE 9. G3 gate: minimum valid population = 22 of 47.

n	bias A (pp)	bias B (pp)	cov% (A/B)	confirm A/B	pass
47	0.3	-3.5	99 / 96	1.5 / -2.2	True
21	1.8	-4.9	96 / 98	2.1 / -5.1	False
22	2.4	-0.8	98 / 99	2.2 / -2.4	True

B. THE TWO REGIMES AND WHAT EACH IMPLIES

The partition θ_P distinguishes two actionable regimes. If the lineage share dominates, cross-family diversification is a credible mitigation: models with independent ancestries add error variance that is not shared, so an ensemble is less likely to fail a question the way its members do. If the era share dominates, diversification alone is insufficient, because models released in the same period share the same training environment regardless of ancestry. The decision rule that follows from RQ6 is intentionally conservative: only a dominant era share refutes diversification-as-remedy for the connected population; a dominant lineage share is a necessary, not sufficient, condition for recommending diversification, because the connected subset may not represent the broader model universe. Independent recent work on ensemble combination reaches the same boundary from the other side: pairwise error

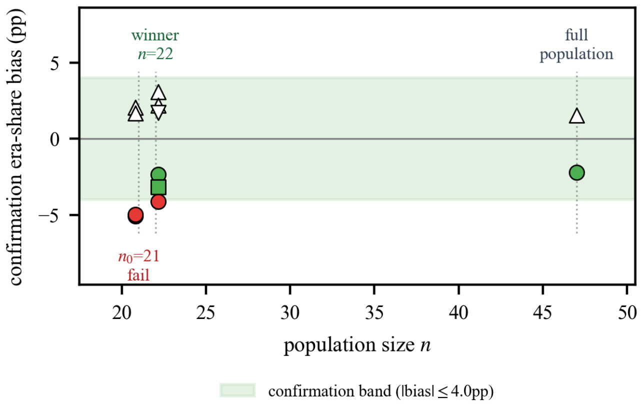

FIGURE 5. Pre-analysis population-design decision trace: population size vs. era-share bias. Markers give the confirmation battery at 1000 repetitions (◯ scenario B, △ scenario A) and the 2000-repetition robustness re-run (□ scenario B, ▽ scenario A); filled markers are within the confirmation band, open markers outside it. Candidates at $n = 21$ and $n = 22$ are offset horizontally about their population size so their marker clusters do not overlap. 47 passes, the structural minimum 21 fails the margin confirmation (knife-edge), and 22 passes — the extra model over the structural minimum is statistically necessary, not computationally convenient. The winner also clears the strict bar at 2000 repetitions (A 1.7pp / B −3.2pp).

correlation substantially *underestimates* the co-failure ceiling of an ensemble, so the operative risk is the joint all-wrong rate [35]; our partition estimates exactly the lineage and era shares whose sum governs that ceiling on the connected population.

C. WHAT A PRACTITIONER DOES DIFFERENTLY UNDER EACH OUTCOME

The two regimes prescribe different procurement behaviour, and the difference is large enough to be worth stating concretely. Under a lineage-dominant outcome, the actionable variable is *ancestry spread*: an operator assembling a three-model panel should spend its budget on models from independent families even at some cost in individual accuracy, because the marginal member contributes error variance that is genuinely unshared. Adding a fourth model from a family already represented buys comparatively little. The audit that follows is a lineage audit: for each candidate, what is it descended from, and is that ancestor already in the portfolio?

Under an era-dominant outcome the same procurement rule is close to useless, because ancestry is not what the models share. The actionable variable becomes *release-date spread*, which is a far less comfortable lever: it means deliberately retaining older models, whose individual accuracy is worse, in order to buy independence, and it means that a portfolio refreshed all at once to the current generation is more correlated than the one it replaced—exactly the opposite of the usual upgrade intuition. It also means the mitigation has a ceiling that diversification cannot raise, and that residual correlated failure must be handled downstream, by human escalation or by out-of-model checks, rather than by portfolio composition.

The practical value of resolving the question is therefore asymmetric. A lineage-dominant answer tells an operator to do something they can do. An era-dominant answer tells them that the intervention they are most likely to reach for does not work, which is less satisfying but considerably more useful than not knowing.

D. STRUCTURAL SAFEGUARDS

Two structural safeguards keep the discussion honest. First, the mechanistic estimand θ_M is reported separately and never merged into θ_P; only the small co-released cohorts and staggered fine-tune chains can hold era exactly fixed, so any mechanistic claim is scoped to them. Second, all conclusions are scoped to the connected subset of the open-model population as defined in Section V-A; the partition is undefined by construction on nested subpopulations, which is a property of the design, not a data gap.

E. LIMITS OF THE RECOMMENDATION

Three limits bound how far any of this can be carried. The population is drawn from open-weight models with documented release records, so conclusions do not transfer to closed models, whose training dates and ancestry are not observable, and whose behaviour may differ systematically precisely because their development is unconstrained by public release conventions. The trait is measured on a single fixed benchmark item set; the shares are shares of *that* trait's variance, and a benchmark weighted differently towards reasoning or recall could reasonably yield a different split. And the family grouping, though verified against organization metadata rather than inferred from behaviour, is a coarse proxy for ancestry: it treats a family as a unit when in reality a family may contain a redesign that shares little with its predecessors—the DeepSeek V4 case documented in Appendix A is exactly this, and is handled by treating it as a new lineage. Each of these is a reason to read the eventual shares as scoped estimates rather than universal constants, and each is revisited in Section VIII.

VIII. THREATS TO VALIDITY, SCOPE RESTRICTIONS, AND LIMITATIONS

The threats below fall into three groups, and the distinction matters because the three call for different responses. The first group are limits the design *detects*: the simulation battery and identifiability audit are constructed so that these announce themselves, and their magnitudes are reported quantities rather than assumptions. The second are limits the design *bounds but cannot remove*: they bias the estimates in a known direction by a quantity we can characterize, and the honest response is to disclose the direction and carry the bound. The third are *scope restrictions*: they are not defects to be mitigated at all, but statements about the population to which any conclusion applies. Presenting all three as a single undifferentiated list of caveats, as is common, obscures the fact that only the second group represents genuine uncertainty about the numbers.

A. LIMITS THE DESIGN DETECTS AND REPORTS

- *Six-family small-sample limit.* The real design has six family levels (df= 5), which caps family-share CI coverage below nominal regardless of CI construction and produces a family-share bias of order −5pp in the lineage-dominant scenario (F2 in the simulation report). Family shares are reported with this limit disclosed; the same limit argues for acquiring more lineage families or reporting family shares as lower bounds.
- *Power and boundary shares.* With $F = 6$ families and 14 quarters, the era trend (RQ5) has limited support and share estimates near the boundary carry inflated CIs; these are reported with the underpower caveat rather than re-aimed.
- *Underpowered binary era.* Item-level binary outcomes cannot resolve the era variance at this occupancy; the continuous per-model trait path is the documented resolution, and any binary-era claim must carry the underpower caveat.

B. LIMITS THE DESIGN BOUNDS BUT CANNOT REMOVE

These are the threats that carry real uncertainty into the estimates. Each has a known sign, which is what makes it reportable rather than merely worrying.

- *Lineage metadata sparsity.* True cross-generation lineage is largely undocumented in model cards; only five cross-generation edges are verified. This constrains the mechanistic estimand, not the primary partition, which uses the verified family grouping. The asymmetry is worth stating plainly: the primary partition is robust to this gap because family membership is documented organizationally, whereas θ_M rests on the edge record and is therefore the weaker of the two estimands by a margin that will not close until `base_model` provenance is more consistently published.
- *Teacher leakage.* Cross-family teacher-student relationships (e.g., Phi-4 from GPT-4o data, Gemma from Gemini) violate the independence assumption to a known-but-unknown degree and are disclosed as belonging to the era channel.
- *Independent trait-error.* Per-model trait measurement errors are assumed independent across models; correlated measurement error would move shares jointly, and is probed by the trait-definition and leaked-drop sensitivity blocks.
- *Common item set.* A fresh five-shot pass on one fixed item set is the basis for all traits; prior work mixing leaderboard subsets is explicitly not used as validation.
- *Benchmark contamination and saturation.* Traits come from a widely cited five-shot benchmark; if pretraining included the items, trait variance is compressed toward zero and era/lineage shares are biased downward. The leaderboard sanity cross-check is explicitly not validation and is so labeled.
- *Temporal drift.* The era component pools 14 quarters under an iid assumption; if the shared training environment is non-stationary within the window, the era effect is misspecified. The era-convergence entry and the connected-window audit probe this assumption.

C. SCOPE RESTRICTIONS

The following are not deficiencies to be mitigated. They define the population to which any estimate applies, and an estimate read outside that population is not a weaker claim but a different and unsupported one.

- *Open-weight sampling frame.* The population is restricted to open-weight models with documented release records; hosted and closed models are excluded by design, so conclusions scope to the connected subset of the open-weight universe and do not generalize to closed systems. This is a consequence of requiring a verified release date and family attribution—the same requirement that makes the design auditable is what excludes closed models.
- *Visibility bias.* The frame is drawn from widely tracked open releases; smaller or niche families are underrepresented, which could overstate the family component relative to the full open-weight universe. Unlike the items above, the direction of this bias is argued rather than measured, and we flag it as the weakest-supported claim in this section.
- *Off-subset scope.* The partition is defined only where lineage and era vary jointly; conclusions do not extend to nested subpopulations (where the partition is undefined by construction) or to closed models. It is worth being explicit that this is not a coverage gap that more data would close: on a nested subpopulation the estimand does not exist, so there is nothing there to estimate more carefully.

IX. CONCLUSION AND FUTURE WORK

We have presented an identifiability-gated variance-decomposition instrument for correlated errors in public language models, and the decision order that makes it defensible: verify identifiability, validate the estimator in simulation, design the study population without observing outcomes, and only then decompose measured traits into lineage, era, and model-specific components on the connected subset. The estimator is validated under both balanced and realistic occupancy and fails detectably under nested mis-specification. A pre-analysis procedure selects a minimum valid population of 22 of 47 models—the smallest population that is structurally identifiable and clears the simulation-based recoverability bar—at about one-third of the full-run cost (a 67% reduction). The empirical partition—the actual lineage and era shares on the selected population—is the outcome of the pre-registered measurement pass and is deliberately not anticipated here; the decision rule for whether cross-family diversification is a credible mitigation for correlated failure is pre-registered and will be evaluated from those shares.

All artifacts—population, design, the optimizer, the validator, and the tests—are released with the paper.

The broader claim we would defend is about order rather than about language models. Three questions—is the quantity identifiable, does the estimator recover it at this sample size, and which units should be measured—are routinely settled after the data are in hand, where each of them becomes a degree of freedom rather than a result. Settling them first costs little: the identifiability argument is algebra, the estimator validation is simulation, and the population design is a search over subsets, none of which requires the measurement it gates. What it buys is that each can be reported as a verdict that a reader can check, including the verdict that the analysis should not proceed. In this instance all three passed, which is the uninteresting outcome; the design's value is that it had a way to return the interesting one.

Three extensions follow naturally. The first is scope: the instrument requires only a documented release date, a family grouping, and a per-model trait, so it applies unchanged to closed models the moment a provider publishes release metadata, and to any trait—latency, refusal rate, calibration error—for which correlated behaviour across a model population is the object of interest. The second is temporal: because the open-model population grows every quarter, the same design can be re-run as a rolling estimate, and the era component becomes a time series rather than a scalar, which is the form in which the convergence question of RQ5 is most naturally asked. The third is resolution: the family grouping is the coarsest defensible ancestry proxy, and as `base_model` provenance becomes more consistently recorded, the lineage factor can be refined from family membership towards the verified edge graph, at which point θ_P and θ_M begin to converge on the same population rather than on disjoint ones.

APPENDIX A PHASE 0 VERIFICATION

Family groupings were verified against Hugging Face organization metadata and technical reports. Release quarter is the public release date, not the HF *createdAt* timestamp; four documented divergences were corrected by hand. DeepSeek V4 is treated as a new independent lineage (a ground-up redesign that dropped the MLA architecture), and the within-family Small chain (Small 3 $\to$ 3.1 $\to$ 3.2 $\to$ 4; Devstral-2 on Small-3.1-Base) is the only verified cross-generation chain.

APPENDIX B FORMAL IDENTIFICATION ARGUMENT

For the covariance model (3), $\mathbf{V}(\theta) = \sigma_U^2\mathbf{I} + \sigma_L^2\mathbf{Z}_F\mathbf{Z}_F^\top + \sigma_E^2\mathbf{Z}_E\mathbf{Z}_E^\top$. REML discards the fixed mean, so θ is identifiable iff the three matrices $\mathbf{I}, \mathbf{Z}_F\mathbf{Z}_F^\top, \mathbf{Z}_E\mathbf{Z}_E^\top$ are linearly independent on the orthogonal complement of $\mathbf{1}$. For $i \neq j$, the (i,j) entry of $\mathbf{V}$ is $b\,\mathbf{1}\{f(i) = f(j)\} + c\,\mathbf{1}\{e(i) = e(j)\}$ with $b = \sigma_L^2$, $c = \sigma_E^2$: a pair sharing only a family pins b, a pair sharing only an era pins c, and a cross pair pins nothing. Failure modes are then only (i) no family-sharing pair with disjoint eras exists—equivalently family and era partitions coincide, the nested design—in which case $\mathbf{Z}_F$ and $\mathbf{Z}_E$ span the same column space and σ_L^2, σ_E^2 are perfectly aliased; and (ii) the family$\times$era incidence graph is disconnected, in which case the constraints separate by component and components alias with a mixture. Every connected crossed design with both marginals active is therefore identifiable. This is the rank condition of Section IV and the gate of Section V; Appendix C probes it empirically.

APPENDIX C DETECTABLE-FAILURE BATTERY (D3)

The nested mis-specification is flagged by three independent detectors, all computed on 300 fresh repetitions of each scenario:

- *BLUP collinearity.* The absolute correlation of family and era BLUPs ($|\operatorname{corr}(\hat{u}_F, \hat{u}_E)|$) exceeds 0.9.
- *SE inflation.* Some variance component has $\widehat{\mathrm{SE}} \geq |\hat{\sigma}|$ (SE at least the estimate).
- *Profile flatness.* The drop in the profile likelihood over a ± 0.4 log-variance window along the aliased direction stays below 1.9207 (the χ_1^2 95% critical value over two).

Non-convergence is tracked as a fourth signal. A design passes the gate only if all three detectors flag the aliasing in $\geq 90\%$ of repetitions and no silent CI coverage is observed; on the final design all three hit 100% and silent coverage is 0%.

APPENDIX D CROSSING IMPLIES IDENTIFIABLE RANK

Let the $N \times (F+E)$ incidence matrix be $\mathbf{C} = [\mathbf{Z}_F\ \mathbf{Z}_E]$. The marginal covariance (3) is $\mathbf{V} = \sigma_U^2\mathbf{I} + \mathbf{C}\mathbf{G}\mathbf{C}^\top$ with $\mathbf{G} = \operatorname{diag}(\sigma_L^2\mathbf{1}_F, \sigma_E^2\mathbf{1}_E)$. The variance map $\theta \mapsto \mathbf{V}(\theta)$ is injective on the orthogonal complement of $\mathbf{1}$ iff $\mathbf{C}$ has full column rank and is not block-reducible. For a connected, crossed design with every family in ≥ 2 eras and every era in ≥ 2 families, the contrast pairs in Appendix B exist for both channels, so the rank condition holds; a nested or disconnected design reduces the effective rank to F (resp. separates components) and the map is non-injective. The Phase 1 design loop (Section V-C) searches exactly over these full-rank crossed subpopulations; the winner (22 of 47) realizes rank 19 of 20.

APPENDIX E REML AND WOODBURY COMPUTATION

The REML objective (4) is maximized over the three log-variances with a numerical optimizer; each evaluation needs $\mathbf{V}^{-1}$ and $\det\mathbf{V}$. Writing $\mathbf{V}^{-1} = \sigma_U^{-2}(\mathbf{I} - \mathbf{C}(\sigma_U^2\mathbf{I} + \mathbf{C}^\top\mathbf{C}\mathbf{G})^{-1}\mathbf{C}^\top)$ (the Woodbury identity, (5)), each likelihood evaluation reduces to linear algebra on $(F+E)\times(F+E) = 20 \times 20$ matrices—$\mathbf{C}^\top\mathbf{C}$ is built once—so a full evaluation is $O(N)$ with negligible constants. Share confidence intervals use a Monte-Carlo scheme on the asymptotic 3×3 covariance of the log-variances, with a PSD clip for sparse designs; the optimizer, the validator, and every reported statistic are released, and the estimator-comparison table (Table 4) is reproducible from the released code under the recorded seeds.

APPENDIX F COMPUTATION AND COST PLAN

All Phase 0 and Phase 1 analyses run on CPU and are fully reproducible. The Phase 2 eval pass runs on one 8$\times$H200-141GB node in fp8; per-model memory and GPU-time plans

TABLE 10. Phase 2 GPU cost plan and access for the 22-model population (cost column: estimated single-GPU minutes).

Family	Model	Quarter	Params	Access	Est. min
Llama	Llama-1	2023Q1	7B	gated	10
Phi	Phi-1	2023Q2	1.3B	public	2
Qwen	Qwen-7B	2023Q3	7B	public	10
Mistral	Mistral-7B	2023Q3	7B	public	10
Phi	Phi-1.5	2023Q3	1.3B	public	2
Phi	Phi-2	2023Q4	2.7B	public	4
Qwen	Qwen1.5	2024Q1	72B	public	102
Phi	Phi-3	2024Q2	14B	public	20
Llama	Llama-3.1	2024Q3	70B	gated	99
Llama	Llama-3.3	2024Q4	70B	gated	99
Phi	Phi-4	2024Q4	3.8B	public	5
Mistral	Mistral-Small-3	2025Q1	24B	gated	34
Mistral	Mistral-Small-3.1	2025Q1	24B	public	34
Phi	Phi-4-reasoning-plus	2025Q2	11B	public	16
Mistral	Mistral-Small-3.2	2025Q2	24B	gated	34
Gemma	Gemma-3n	2025Q2	4B	gated	6
DeepSeek	DeepSeek-V3.1	2025Q3	671B/37B	public	53
Mistral	Devstral-2	2025Q4	24B	gated	34
DeepSeek	DeepSeek-V3.2	2025Q4	685B/37B	public	53
Phi	Phi-4-reasoning-vision-15B	2026Q1	15B	public	21
Mistral	Mistral-Small-4	2026Q1	32B	gated	45
Gemma	Gemma-4-12B	2026Q2	12B	public	17
22-model total					710
47-model full roster					2,154

are released with the artifacts. Table 10 is the cost plan for the selected 22-model population: 710 estimated single-GPU minutes against 2,154 for the full 47-model roster—about one-third of the full-run cost (a 67% reduction). Eight of the 22 are gated and require license acceptance.

APPENDIX G MINIMUM VALID POPULATION ROSTER

Table 11 records, for each of the 22 selected models, the single binding reason for inclusion, matching the decision log. Fourteen are public; eight are gated. Excluded models fail an era-window or a structural-identifiability requirement, or sit off the connected subset. A model is *era-window-forced* when removing it would strand an era window with no usable model; *edge/chain-forced* when it is the unique carrier of a verified lineage edge or of the Mistral Small chain (the mechanistic estimand's only cross-generation support); and *crossing/rank-VIF-forced* when it is required for a crossed full-rank design with VIF ≤ 10.

TABLE 11. Roster of the minimum valid population with the binding inclusion reason.

Family	Model	Quarter	Params	Access	Reason
Llama	Llama-1	2023Q1	7B	gated	era-window
Phi	Phi-1	2023Q2	1.3B	public	era-window
Qwen	Qwen-7B	2023Q3	7B	public	crossing
Mistral	Mistral-7B	2023Q3	7B	public	rank/VIF
Phi	Phi-1.5	2023Q3	1.3B	public	rank/VIF
Phi	Phi-2	2023Q4	2.7B	public	era-window
Qwen	Qwen1.5	2024Q1	72B	public	era-window
Phi	Phi-3	2024Q2	14B	public	era-window
Llama	Llama-3.1	2024Q3	70B	gated	edge/chain
Llama	Llama-3.3	2024Q4	70B	gated	edge/chain
Phi	Phi-4	2024Q4	3.8B	public	edge/chain
Mistral	Mistral-Small-3	2025Q1	24B	gated	edge/chain
Mistral	Mistral-Small-3.1	2025Q1	24B	public	edge/chain
Phi	Phi-4-reasoning-plus	2025Q2	11B	public	edge/chain
Mistral	Mistral-Small-3.2	2025Q2	24B	gated	edge/chain
Gemma	Gemma-3n	2025Q2	4B	gated	crossing
DeepSeek	DeepSeek-V3.1	2025Q3	671B/37B	public	era-window
Mistral	Devstral-2	2025Q4	24B	gated	edge/chain
DeepSeek	DeepSeek-V3.2	2025Q4	685B/37B	public	edge/chain
Phi	Phi-4-reasoning-vision-15B	2026Q1	15B	public	edge/chain
Mistral	Mistral-Small-4	2026Q1	32B	gated	edge/chain
Gemma	Gemma-4-12B	2026Q2	12B	public	era-window

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